ASSIGNMENT 1

1. In 2 two sentences, explain quota sampling technique
2. How can you identify outliers using the IQR method
3. A retail store recorded the number of products sold over 20 consecutive days; 25, 30, 28, 35,40, 45, 38, 36, 42, 39, 41, 44, 37, 33, 31, 29, 34, 46, 43, 40, (a) Calculate the mean, median, mode (b) Compute the range, variance and standard deviation (c) Determine the first quartile (Q1), second quartile(Q2), and third quartile(Q3). (d) Interpret the results and describe the distribution of sales.

ANSWERS:

1. Quota sampling: This is a non-probability sampling method whereby the researcher divides the population into subgroups and sets a specific quota for how many people to sample from each group. Then researchers will select participants conveniently until each quota is filled without random selection.
2. To identify outliers using IQR method: First, you calculate Q1(25^{th} percentile) and Q3 (75^{th} percentile), then find IQR = Q3 – Q1.

25, 28, 29, 30, 31, 33, 34, 35,36, 37, 38, 39, 40, 40, 41, 42, 43, 44, 45, 46

Q2 = median = second quartile, number = 20 values

Median = 37 + 38 /2 = 37.5

Q1 = median of upper half = 25, 28, 29, 30, 31, 33, 34, 35, 36, 37

Q1 = 31 + 33 / 2 = 32.0

Q3 = median of lower half = 38, 39, 40, 40, 41, 42, 43, 44, 45, 46

Q3 = 41 + 42 /2 = 41.5

Using the IQR method = Q3 – Q1

41.5 – 32.0 = 9.5

For the outliers boundaries:

Lower bound = Q1 – 1.5 * IQR

Upper bound = Q3 – 1.5 * IQR

Lower bound = 32.0 – 1.5 * 9.5 = 17.75

Upper bound = 41.5 + 1.5 * 9.5 = 55.75

Therefore, there is no outliers in this data

3. Mean = 736 / 20 = 36.8

Median = 37.5

Mode = 40(it appears twice)

Range, variance and standard deviation

Range = Highest number – Lowest number

Range = 46 – 25 = 21

Variance = 751.20 / 2 = 37.56

Standard Deviation = $\sqrt{37.56 = 6.13}$

Q1 = 32.0

Q2 = 37.5

Q3 = 41.5

Interpretation of the results:

The mean 36.8 is close to the median 37.5, which means the distribution is roughly symmetric with no strong skew.

The standard deviation is about 6.13 shows that daily sales varied moderately around the average.

The most common number of products sold was 40.

There are no outliers in the data.